

GLOSSARY

API Gravity A scale (API or American Petroleum Institute scale) commonly used in the petroleum industry to characterize the quality of a crude oil. It is defined as follows

$$\text{API gravity} \equiv \frac{141.5}{\rho} - 131.5,$$

where ρ is the density of the oil in gm/cm^3 .

BRD The Butler-Reeds-Dawson method [48] for solving the Fredholm equation of the first kind with a solution commensurate with the error bar of the data.

Bulk Volume Irreducible The part of the porous earth formation occupied by immobile, capillary bound water as well as the clay bound water. It is also called the irreducible water.

BVI See Bulk Volume Irreducible.

Carr-Purcell-Meiboom-Gill Pulse Sequence A pulse sequence used to measure T_2 relaxation time. It was first suggested by Carr and Purcell [54] and later modified by Meiboom and Gill [161]. The pulse sequence consists of an initial $\pi/2$ pulse and successive π pulses, where the successive π pulses are separated by 2τ and are excited at time $\tau, 3\tau, 5\tau, \dots$. The spin echoes occur at time $2\tau, 4\tau, 6\tau, \dots$, etc..

CMRTM¹ The Schlumberger Combinable Magnetic Resonance logging tool.

CPMG See Carr-Purcell-Meiboom-Gill Pulse Sequence.

CVM Constituent Viscosity Model, a phenomenological model suggested by Freedman et al. [90] to link the relationships between relaxation time vs viscosity and diffusion coefficient vs viscosity for crude oils.

Differential Spectrum Method A hydrocarbon typing method which uses different repeat delays and the large T_1 contrasts between gas, oil, and water to detect the presence and estimate the quantity of hydrocarbons.

¹Trademark of Schlumberger Corporation

DSM See Differential Spectrum Method.

EDM See Enhanced Diffusion Method.

Enhanced Diffusion Method A hydrocarbon typing method which uses a single large TE and two different wait times, one allows full polarization and another allows partial polarization of the hydrogen nuclei.

FFI See Free Fluid Index.

FID See Free Induction Decay.

Formation Factor The ratio of the conductivity of a fluid to that of a porous medium saturated by the same fluid, where the conductivity of the solid matrix is assumed to be zero.

Free Fluid Index The part of the porous earth formation occupied by mobile water or producible fluids.

Free Induction Decay It is the signal detected in the rf coil immediately after the a single rf pulse, usually a 90° pulse.

GOR See Gas Oil Ratio.

Gas Oil Ratio The Gas Oil Ratio (GOR) is defined as

$$\text{GOR} = \frac{\text{cubic meters of methane at standard condition}}{\text{cubic meters of stock tank oil}},$$

where the standard condition is 60°F and 1 atmospheric pressure.

Gyromagnetic Ratio The ratio of magnetic moment μ to the spin angular momentum $\mathbf{J}$, denoted by γ . For protons, $\gamma_p = 2.675197 \times 10^8 \text{ rad s}^{-1} \text{ Tesla}^{-1}$ (or $\times 10^4 \text{ rad s}^{-1} \text{ Gauss}^{-1}$), or $\gamma_p/2\pi = 4.257707 \times 10^7 \text{ Hz/Tesla}$ (or $\times 10^3 \text{ Hz/Gauss}$).

HI See Hydrogen Index.

Hydrocarbon Typing Methods or procedures for characterizing the hydrocarbon types, i.e., gas, or oil (high or low viscosity).

Hydrogen Index It is defined as the amount of hydrogen per unit volume of a fluid divided by the amount of hydrogen per unit volume of pure water in standard conditions.

Irreducible Water Saturation The portion of the water in the porous formation that cannot be produced under a certain capillary pressure.

Larmor Frequency The frequency at which nuclear spins precess about the applied static magnetic field. It is given by $f = \gamma H / (2\pi)$, where γ is the gyromagnetic ratio and H is the applied magnetic field.

Longitudinal Relaxation Time Also called Spin-Lattice Relaxation Time. It is a time constant, denoted as T_1 , which characterizes the relaxation process of energy transfer between the magnetic energy of the spin system and the thermal energy of its surroundings. If the static magnetic field is along the z direction and the equilibrium magnetization is M_0 , Bloch introduced the assumption that the change of M_z with time can be described by the following equations

$$\frac{dM_z}{dt} = \frac{M_0 - M_z}{T_1}.$$

Magnetization The vector sum of magnetic moments per unit volume.

MAS Magic Angle Spinning, a technique routinely used in solid state NMR to average out chemical shift anisotropy and internuclear dipolar interactions.

MRIL^{TM 2} The Halliburton/Numar's Magnetic Resonance Imaging Logging tool.

MRF Magnetic Resonance Fluid characterization method. It is a method developed by Schlumberger for characterizing formation fluids using information obtained from multiple NMR logging measurements.

NML^{TM 3} The Nuclear Magnetism Logging tools, operated by the free precession of spins in the earth's magnetic field. Operation of this tool required special doping of the borehole fluid. It has become obsolete after the introduction of modern pulsed NMR logging tools in the 1990's.

NMR See Nuclear Magnetic Resonance.

²Trademark of Halliburton/Numar Corporation

³Trademark of Schlumberger Corporation

Nuclear Magnetic Resonance A physical phenomenon discovered by Bloch and Purcell independently in 1946, for which both of them were awarded Nobel Prize in 1952. It is a phenomenon found in systems of nuclei that possess both magnetic moments and angular momentum. At resonance condition, radio frequency excitations, tuned to the characteristic frequency of the magnetic system, i.e., the Larmor frequency, are used to stimulate the magnetic nuclei which absorb and re-emit energy through interactions with other nuclei undergoing thermal motions.

OBM Oil Based Mud.

OBMF Oil Based Mud Filtrate.

PAPs See Phase-Alternated Pairs.

PFGSE See Pulsed Field Gradient Spin Echo.

Phase-Alternated Pairs Two CPMG echo trains, the second one is acquired with a phase which differs from that of the first one by 180° . The two echo trains are combined to remove the effects of ringing and baseline offset.

Porosity Unit Porosity of a rock is usually expressed in volume fractions, e.g., 0.25, or in percentage, i.e., 25%, which means 25% of the bulk volume is pore space. Porosity unit is a unit commonly used in the petroleum industry, which is numerically the same as the percentage, e.g., 25 p.u. represents a porosity of 0.25 in volume fraction or 25% in percentage.

p.u. See porosity unit.

Pulsed Field Gradient Spin Echo A pulsed field gradient technique frequently used for determining the self diffusion coefficient of fluids and for studying the physical properties of porous media. It consists of field gradient pulses separated by various lengths of diffusion time.

RD See Repeat Delay.

Repeat Delay Between successive π pulse excitations in a CPMG experiment, a time period is inserted to ensure all spins fully return to their thermal equilibrium state along the direction of the static magnetic field

and are ready for the next excitation. This time period is referred to as "Repeat Delay", "Wait Time", or "Repeat Time" (and abbreviated as RD, WT, TW, or TR in the literature).

Restricted Diffusion The diffusion of particles, which occurs within the pore space of a porous medium, is restricted by the presence of pore walls, and has a particle density profile different from that in an infinite medium, hence the term, restricted diffusion.

Return To Origin Probability The probability of a diffusing particle to return to its original starting point, i.e., the origin, after a certain length of diffusion time.

rf pulse Radio frequency pulse, which has a carrier frequency equal to the Larmor frequency of the spins and a width (or duration) determined by the experimental condition.

RTOP See Return To Origin Probability.

Shift Spectrum Method A method for hydrocarbon typing that uses different TEs to differentiate oil and water in the T_2 distributions.

Singular Value Decomposition A method in linear algebra for solving simultaneous linear equations.

Spin Echo A refocusing of the dephased spins due to field inhomogeneity in the plane perpendicular to the applied static magnetic field, first discovered by Hahn [96] in 1950.

Spin-Lattice Relaxation Time See Longitudinal Relaxation Time.

Spin-Spin Relaxation time See Transverse Relaxation Time.

SSM See Shift Spectrum Method.

Stimulated Echo Stimulated echo is the echo occurred at $\tau_1 + \tau_2$ after three 90° pulses. It was discussed in Hahn's paper on spin echoes [96]. There are additional echoes at $2\tau_2 - 2\tau_1$, $2\tau_2 - \tau_1$, and $2\tau_2$ when $2\tau_1 < \tau_2 < T_2$. For $\tau_1 < \tau_2 < 2\tau_1$, the signal at $2\tau_2 - 2\tau_1$ is missing, but the other remain. Because of the large field inhomogeneities in the probed zone of the logging tools, stimulated echoes occurred in CPMG echo trains

in the logging measurements, and corrections are required to remove this effect.

Surface Relaxivity A quantity that measures the strength of the solid/fluid interface of a porous medium on how fast it can relax the spins in the pore fluid. It has a unit of length/time.

SVD See Singular Value Decomposition method.

T_1 See Longitudinal Relaxation Time.

T_2 See Transverse Relaxation Time.

T_2^* The decay time constant of a Free Induction Decay.

$T_{2,LM}$ The logarithmic average of the T_2 relaxation times.

TDA See Time Domain Analysis.

Time Domain Analysis Subtract the echo train of short wait time from the echo train of long wait time and invert the difference of the two echo trains to obtain a T_2 distribution which reflects the signal due to the difference in wait times. This analysis is called Time Domain Analysis.

TE Time between successive echoes in a CPMG echo train. It is equal to 2τ , where τ is the time between the initial 90° rf pulse and the subsequent 180° pulse.

Transverse Relaxation Time Also called Spin-Spin Relaxation Time. It is a time constant, denoted as T_2 , which characterizes the spin relaxation process in the plane perpendicular to the applied static magnetic field direction. If the static magnetic field is along the z direction, the decay of the magnetization in the xy plane (i.e., M_x and M_y) is described by the following Bloch equations

$$\frac{dM_x}{dt} = -\frac{M_x}{T_2}, \quad \frac{dM_y}{dt} = -\frac{M_y}{T_2}.$$

TR See Repeat Delay.

TW See Repeat Delay.

UPEN Uniform Penalty method, an inversion method for NMR data processing suggested by Brown and his colleagues [34] which provides an uniform penalty along the relaxation time axis.

WT See Wait Time.

Wait Time See Repeat Delay.

Wettability It is defined as the tendency of one fluid to spread on or adhere to a solid surface in the presence of other immiscible fluids.

ϕ Notation used in petrophysics to denote the volume fraction of the pore space. $\phi \equiv V_p/V$, where V_p and V are the pore volume and total volume, respectively.